

Why might AI adversely affect exposed entry-level workers more than other age groups? One possibility is that AI disproportionately substitutes for workers using codified knowledge, including both the “book-learning” that forms the core of formal education and the insights in digital company data that can be codified by AI. AI may be less capable of replacing tacit knowledge, the idiosyncratic tips and tricks that accumulate with experience but which are never digitized.⁴ As young workers supply relatively more codified knowledge than tacit knowledge, they may face greater task replacement from AI in exposed occupations, leading to greater employment reallocation (Acemoglu and Autor, 2011). Older workers with accumulated tacit knowledge may face less task replacement. These benefits of uncoded knowledge may accrue less to non-college workers in occupations with low returns to experience. In other words, AI may be automating the *codifiable*, *checkable* tasks that historically justified entry-level headcount, while complementing the judgment-, client-, and process-intensive tasks performed by experienced workers.

Other explanations may contribute. AI may raise the leverage of experienced staff, increasing their effective span of control (Ide, 2025). Reduced hiring may also be the lowest-friction adjustment margin, compounded by inefficient incentives to train entry-level workers who may move firms (Becker, 1994; Garicano, 2025; Garicano and Rayo, 2025). Consequently firms may primarily shrink junior inflows rather than displace incumbents.

This combination of task substitution at the apprentice margin, complementarity at the expert margin, and quantity (not price) adjustment under wage ladders can explain why early-career employment falls while employment of older workers continues to grow. An important direction for research is to further model and test these predictions.

2 Related Literature

A growing body of research has sought to measure the employment effects of AI.⁵ This includes influential papers that established methodologies for estimating which occupations and tasks were susceptible to automation (Frey and Osborne, 2017; Brynjolfsson and Mitchell, 2017; Brynjolfsson et al., 2018; Felten et al., 2018, 2019; Webb, 2019; Felten et al., 2021). More recently, work such

⁴Ironically, one of the practical skills more likely to be learned on the job than in university computer science classes may be how to use AI for software development.

⁵There has also been extensive media discussion of this topic including Thompson (2025); Raman (2025); Roose (2025); The Economist (2025); Frick (2025). See Online Appendix B for further references and discussion.